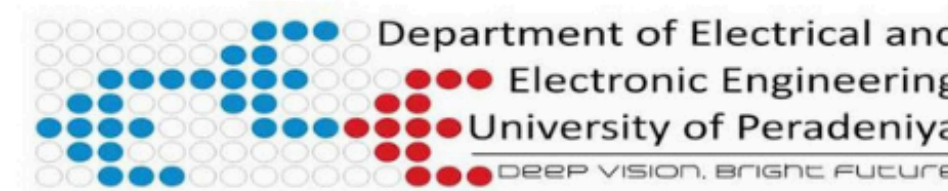




DUAL-ARCHITECTURE IOT TELEMETRY SYSTEM

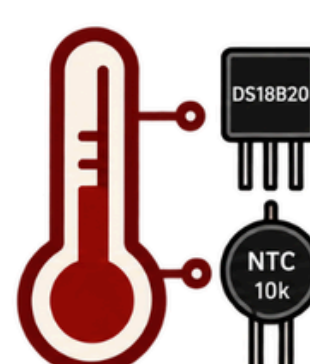


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1. PROJECT OVERVIEW

A Dual-Sensor IoT Telemetry System combines a **DS18B20 Digital Temperature Sensor** and a **10 kΩ NTC Thermistor** to achieve accurate and responsive temperature monitoring. Sensor data is processed at the edge using an adaptive sensor-fusion algorithm and transmitted through a LoRa wireless link to the central telemetry system, where the data is forwarded to MQTT and visualized through Ignition SCADA.



SENSE
Dual-Sensor
Temperature
Measurement



FUSE
Edge Processing
and Calibration



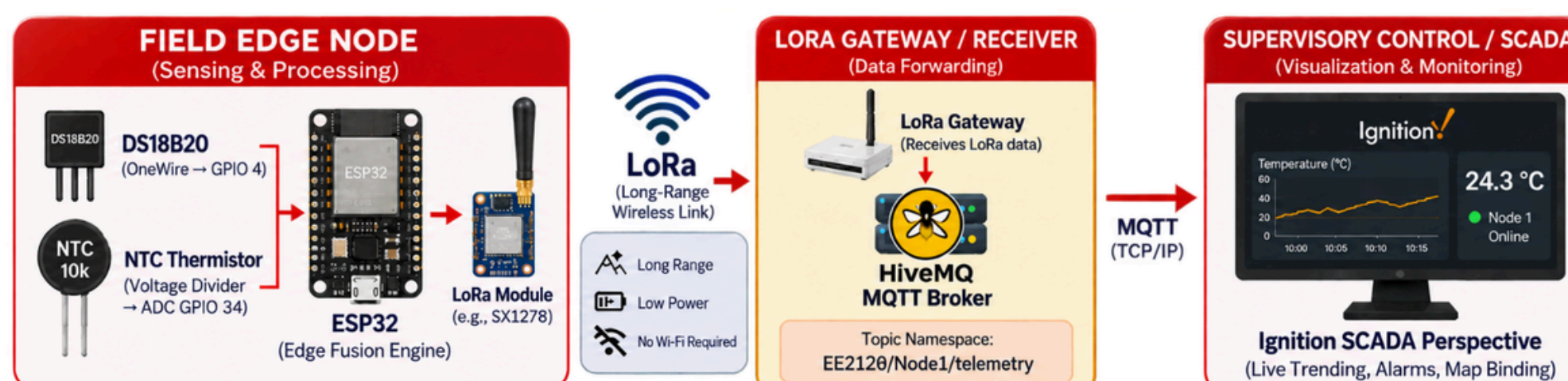
TRANSMIT
LoRa Wireless
Communication



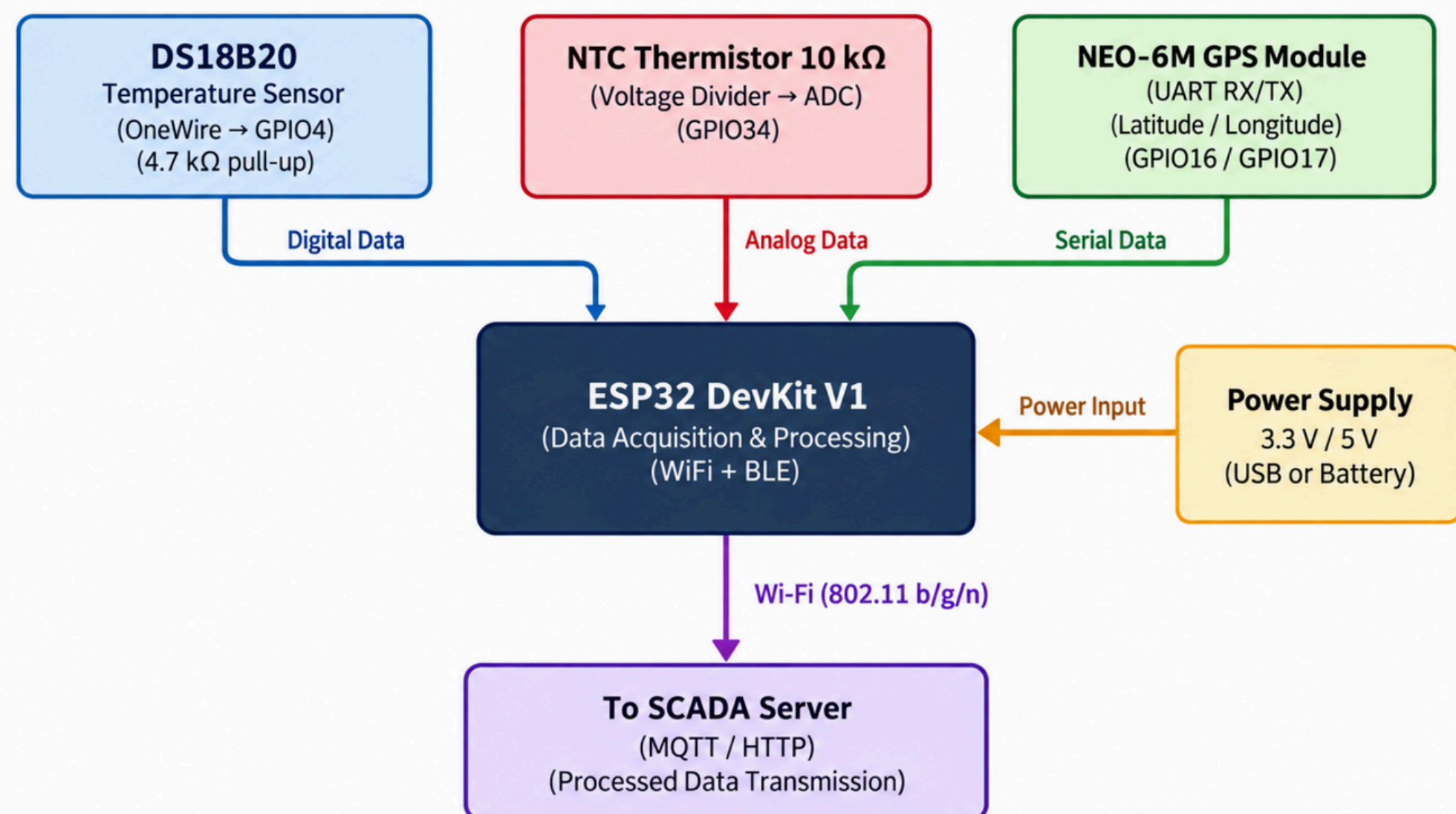
MONITOR
Real-time
SCADA
visualization

2. SYSTEM ARCHITECTURE

The system follows a three-tier IoT architecture that integrates dual-sensor temperature measurement, edge-based data processing, LoRa wireless communication, and real-time SCADA monitoring. The ESP32 acts as the edge telemetry node, acquiring and processing data from the DS18B20 and NTC sensors before generating the fused temperature data. The processed data is transmitted through LoRa to the gateway/receiver and then forwarded to the HiveMQ MQTT broker. The telemetry data is subsequently integrated into the Ignition SCADA platform for continuous visualization, monitoring, and alarm management.

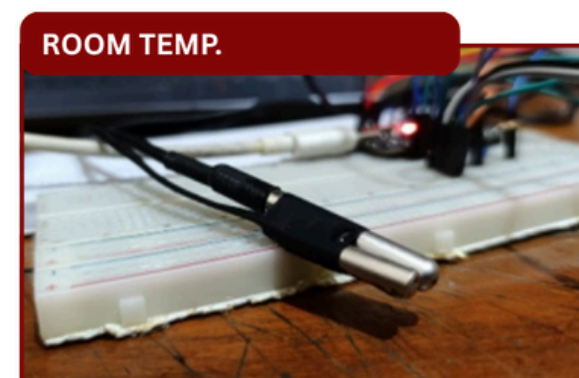
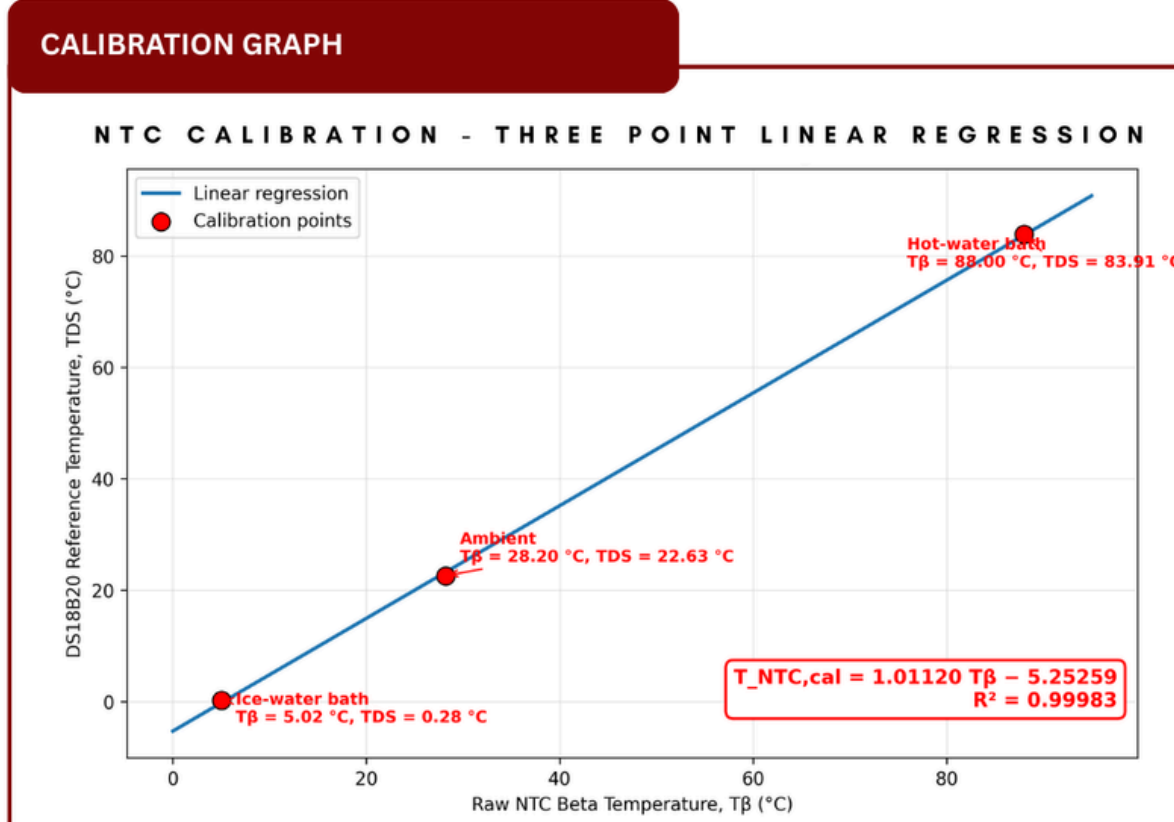


3. HARDWARE DESIGN



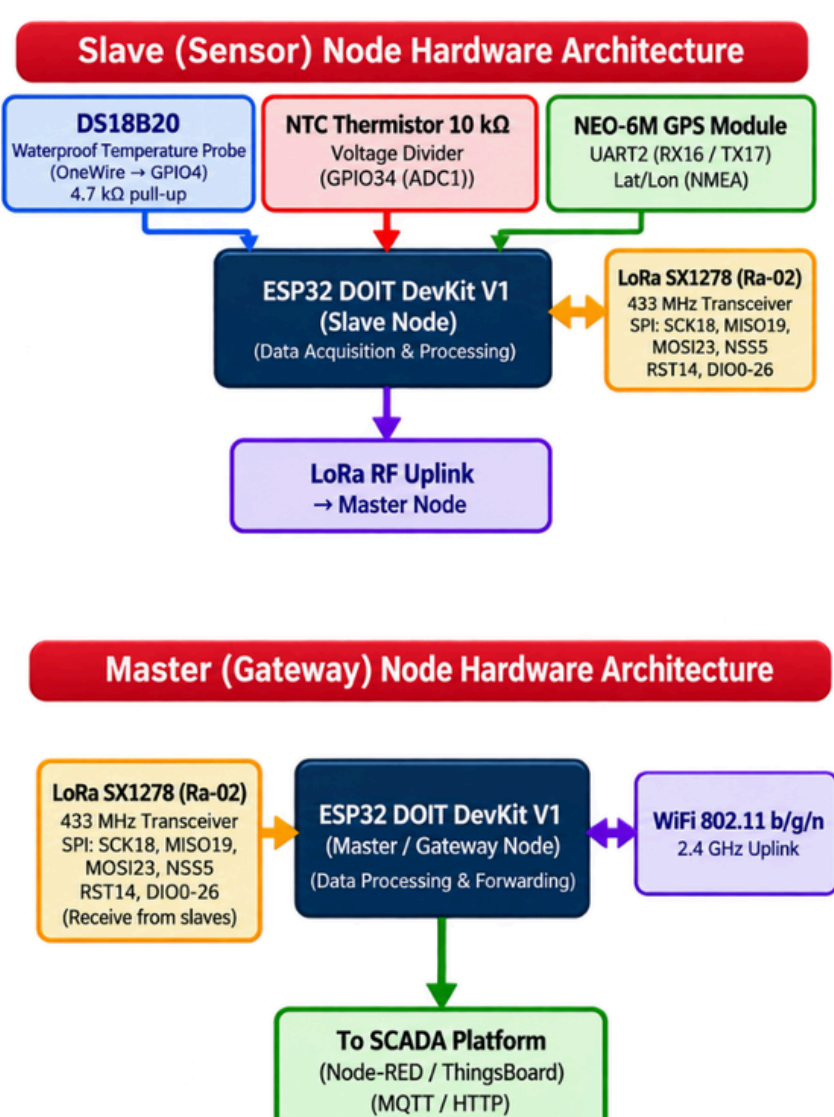
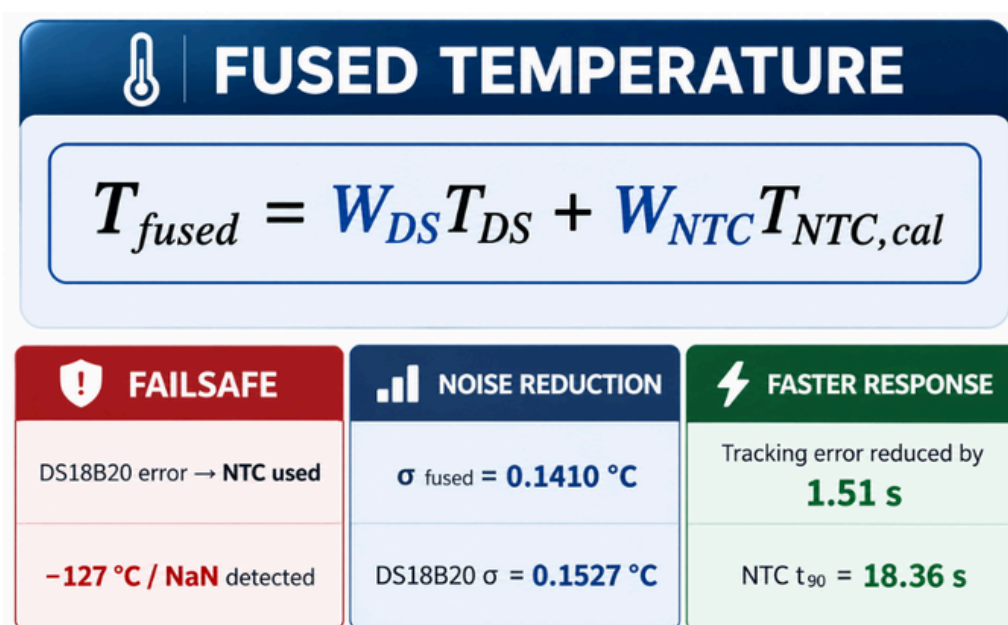
4. CALIBRATION RESULTS

The NTC thermistor was calibrated using the DS18B20 as the reference sensor at three known temperature points: ice-water bath ($\approx 0^\circ\text{C}$), heated air (mid-range), and hot-water bath (high temperature). The measured data was used to obtain a linear calibration model between the raw NTC Beta temperature and the reference temperature.

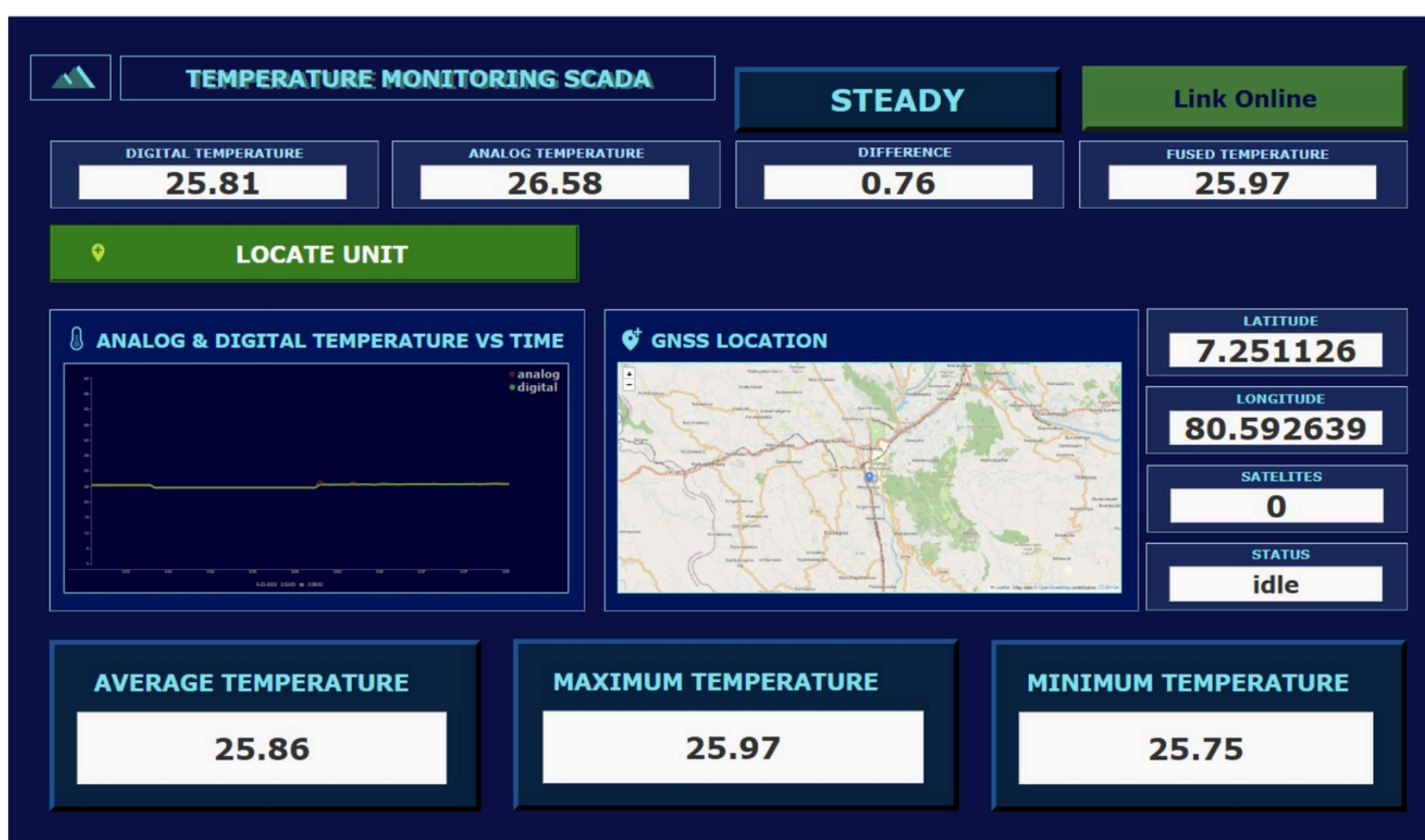


5. SENSOR FUSION LOGIC

The adaptive sensor-fusion algorithm combines DS18B20 and calibrated NTC measurements to produce a reliable fused temperature. The weighting changes according to the temperature difference, giving greater weight to the NTC during transient conditions. A failsafe mechanism and adaptive weighting improve measurement stability and response speed.



6. RESULTS & SCAADA VISUALIZATION



7. CONCLUSION

The dual-architecture IoT telemetry system combines the precision of the DS18B20 digital sensor with the faster thermal response of the 10 kΩ NTC thermistor. The NTC is calibrated using the DS18B20 as a reference, while an adaptive sensor-fusion algorithm dynamically adjusts the sensor weights according to the operating condition. A failsafe mechanism maintains reliable temperature estimation when the DS18B20 becomes unavailable.

The processed fused temperature data is transmitted through a LoRa wireless link to the receiving gateway and forwarded to the HiveMQ MQTT broker. The telemetry is then integrated into the Ignition SCADA platform for real-time temperature visualization, monitoring, and alarm management. Experimental results demonstrate improved measurement stability, transient response, and fault resilience, making the system suitable for continuous IoT-based temperature monitoring.